Alexander Bui

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Education

Texas A&M University College Station, TX

Bachelor's of Science, Computer Engineering, GPA: 4.0 *May 2028*

- Craig and Galen Brown Engineering Honors, Dean's Honor Roll
- Relevant Coursework: Data Structures and Algorithms, Computer Architecture, Intro to Computer Systems, Program Design and Concepts, Foundations of Software Engineering, Software Reverse Engineering

Work Experience

Software Engineer Intern, American Airlines May 2026 — Aug 2026

- Took ownership of 25+ user stories across 9 Agile sprints, engineering over 40 full-stack PRs to power a mission-critical platform utilized daily to assign hundreds of flight sequences.
- Fortified backend optimization workflows by integrating new business constraints into a FICO Xpress solver built on Spring Boot, ensuring automated pilot assignments strictly adhered to FAA regulations.
- Spearheaded the integration of a robust Playwright end-to-end testing framework to validate critical scheduling workflows, reducing deployment risk and saving an estimated 10+ hours of manual QA testing per sprint cycle.

AI/ML Software Engineer, Texas Cyber Range Jun 2025 — Dec 2025

- Orchestrated agentic AI workflows with LangGraph to automate customer support, integrating RAG pipelines and Qdrant vector stores for real-time context retrieval, reducing average ticket resolution time by 500%.
- Designed a containerized microservices architecture deployed on OpenShift (Kubernetes), utilizing Docker and CI/CD pipelines for automated container builds and scaling.
- Built a custom React dashboard for human-in-the-loop agent oversight, enabling support staff to review AI drafts.

Student Technician, Texas A&M University Aug 2025 — Present

- Triaged cybersecurity incidents across a large-scale academic network, utilizing Elastic to correlate network logs and identify Indicators of Compromise (IOCs), mitigating account takeovers and MFA bypass attempts.

Undergraduate Teaching Assistant, Texas A&M University Aug 2025 — Dec 2025

- Enhanced student learning for a class of 30 students by clarifying topics from coding lectures and conducting weekly office hours to assist students one-on-one with debugging Python lab assignments and understanding software concepts.

Software Engineer Intern, GravityIntel Jan 2025 — Aug 2025

- Engineered a full-stack application leveraging agentic AI to automate the analysis, management, and evaluation of government contract bids.
- Architected a hybrid data solution using PostgreSQL for relational data and Couchbase for NoSQL and high-performance vector search to manage complex bid information.

Projects

Aggie Events | *React, NextJS, ExpressJS, PostgreSQL, Docker* Sep 2024 — Present

- Engineered a secure Node.js/Express REST API with PostgreSQL and OAuth 2.0 authentication.
- Enhanced website deployment speeds by 250% through a custom CI/CD pipeline using Docker and Bash.
- Streamlined tasks for a 27-member team using Agile methodologies, earning **Best Project Manager** for Fall 2024.

OpenGL Graphics Engine | *C++, OpenGL, CMake* Apr 2023 — Present

- Built a graphics pipeline from scratch using OpenGL and C++, implementing rendering with an Entity-Component-System (ECS) architecture, supporting lighting, textures, and custom model loading.

Skills

- **Technologies**: C/C++, JavaScript, Python, SQL, Java, Spring Boot, TypeScript, React, Angular, NextJS, Playwright, Node.js, PostgreSQL, Docker, CUDA, Linux, Bash
- **Awards**: 1st TAMUHack 2025 (Toyota), 2nd TAMU Datathon 2025 (Hitachi), 2nd TAMUHack 2025 (AA), 3rd Datathon 2024 (TAMIDS), Cyberpatriot National Platinum Semifinalist